

ECEN-662: Homework 5

Due on March 8, 2026 at 11:59 PM

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Problem 1

Let $X = (X_1, X_2)$ where

$$X_i \stackrel{iid}{\sim} \text{Exp}(\theta)$$

- (a) Find the UMVUE for $1/\theta$.
- (b) Show that the geometric mean $\sqrt{X_1 X_2}$ has a smaller MSE than the UMVUE for all $\theta > 0$.
- (c) Find another estimator that has an even smaller MSE for all $\theta > 0$.

Solution

Part (a):

Part (b):

Part (c):

Problem 2

Let $X = (X_1, \dots, X_n)$ where

$$X_i \stackrel{iid}{\sim} f_\theta(x_i) = \exp(-(x_i - \theta))1_{\{x_i \geq \theta\}}.$$

- (a) Find the MLE of θ and calculate its bias and MSE.
- (b) Find the UMREE of θ under the squared-error loss and calculate its bias and MSE. How do you compare them with those of the MLE?

Solution

Part (a):

Part (b):